

TMA: L^AT_EX101-sample

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Update for 2025/09/30 v1.21

This document is provided as an example of how to use the `ou-tma` package. With the 7th April 2025 version of the package there has been an update to the name of the package. Gone is ~~`tma`~~ and in comes `ou-tma`. The reason for the change is twofold. Firstly there are many internal changes to make the package more in line with the conventions of the L^AT_EX3 project, and secondly the introduction of the package into the CTAN archive.

In presenting to the CTAN archive, the regular stylesheet has been wrapped up in a L^AT_EX document file—a `.dtx` file. The beauty of the `.dtx` file is that the `.sty` and source for it's documentation ship together in a single file. A second simple `.ins` file is also needed to extract the `.sty`, but that is unlikely to change and can be created simply on the fly if needed.

To use the options start your document with:

```
\documentclass[a4paper,12pt]{article}
\usepackage[OPTION]{ou-tma}

\myname{...}
\mypin{...}
\mycourse{...}
\mytma{...}
\date{...}
\begin{document}
\maketitle
...
```

Where `OPTION` is one of the following

- `[roman]`
- `[alph]` *Default*
- `[cleveref]`
- `[pdfbookmark]`
- `[legacy]`

See the `ou-tma.pdf` file for further details of available options.

As we begin this example, we can use a new command `\setquestionstring` to change the default ‘Q 1’, ‘Q 2’, etc to something else. In this case we will use `\setquestionstring{Question}` to make the questions separators display as ‘Question 1’, ‘Question 2’, etc.

Question 1

- (i) We have $1 = 10^0$ and $1 + 2 + 3 + 4 = 10^1$. Prove that there are no other powers of ten which are the sum of the first n integers.

We have:

$$\sum_{i=1}^n i = \frac{(n)(n+1)}{2}$$

Let

$$\begin{aligned} \frac{(n)(n+1)}{2} &= 10^x \\ \Rightarrow (n)(n+1) &= 2^{x+1}5^x \end{aligned}$$

Now, either n is odd, or $n+1$ is odd.

Consider the case where n is odd:

By the Fundamental Theorem of Arithmetic, n can only have the prime factors 2 or 5. Since it is odd, it can only be a perfect power of 5. Now, $n+1$ also can only have the prime factors of 2 or 5. If n is divisible by 5, then $n+1$ is not divisible by 5. Therefore $n+1$ is a perfect power of 2. Therefore:

$$\begin{aligned} n &= 5^x & \text{and} & & n+1 &= 2^{x+1} \\ & & & & \Rightarrow x &= 0 \end{aligned}$$

(for any higher x , $5^x \gg 2^{x+1}$)

$$\Rightarrow n = 1$$

Now consider the case where $n+1$ is odd:

By similar arguments to above, $n+1$ must be a perfect power of 5 and n must be a perfect power of 2.

$$\begin{aligned} n &= 2^{x+1} & \text{and} & & n+1 &= 5^x \\ & & & & \Rightarrow x &= 1 \end{aligned}$$

(for any higher x , $5^x \gg 2^{x+1}$)

$$\Rightarrow n = 4$$

Therefore $n = 1$ and $n = 4$ are the only solutions to the original problem. ☹

(iii) (a) Show that:

$$\sum_{x=1}^n x(x+1) = \frac{n(n+1)(n+2)}{3}$$

Let

$$f(n) = \frac{n(n+1)(n+2)}{3}$$

Now, adding the $n+1$ term to the above

$$\begin{aligned} f(n) + (n+1)(n+2) &= \frac{n(n+1)(n+2)}{3} + (n+1)(n+2) \\ &= \frac{(n^3 + 3n^2 + 2n+)}{3} + n^2 + 3n + 2 \\ &= \frac{(n^3 + 6n^2 + 11n + 6)}{3} \\ &= \frac{((n+1)(n+2)(n+3))}{3} \\ &= f(n+1) \end{aligned}$$

Therefore, if $f(n)$ is valid, then so is $f(n+1)$.

Since $1 \times 2 = 2 = \frac{1 \times 2 \times 3}{3} = f(1)$, then $f(n)$ is valid for all $n \geq 1$. ☹

(b) (iii)(b) Show that:

$$\sum_{x=1}^n x^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

Let

$$f(n) = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

Now, adding the $n+1$ term to the above

$$\begin{aligned} f(n) + (n+1)^4 &= \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} + (n+1)^4 \\ &= \frac{1}{30}(6n^5 + 15n^4 + 10n^3 - n) + (n^4 + 4n^3 + 6n^2 + 4n + 1) \\ &= \frac{1}{30}(6n^5 + 15n^4 + 10n^3 - n + 30n^4 + 120n^3 + 180n^2 + 120n + 30) \\ &= \frac{1}{30}(6n^5 + 45n^4 + 130n^3 + 180n^2 + 119n + 30) \end{aligned} \quad (1.1)$$

Now,

$$\begin{aligned} f(n+1) &= \frac{(n+1)((n+1)+1)(2(n+1)+1)(3(n+1)^2+3(n+1)-1)}{30} \\ &= \frac{1}{30}(n+1)(n+2)(2n+3)(3n^2+9n+5) \\ &= \frac{1}{30}(6n^5 + 45n^4 + 130n^3 + 180n^2 + 119n + 30) \end{aligned} \quad (1.2)$$

Comparing equation (1.1) with equation (1.2) we see that

$$f(n) + (n+1)^4 = f(n+1)$$

Therefore, if $f(n)$ is valid, then so is $f(n+1)$.

Since $1^4 = 1 = \frac{1 \times 2 \times 3 \times 5}{30} = f(1)$, then $f(n)$ is valid for all $n \geq 1$. ☺

Note that by using
`\begin{question}[3]`
 at this point, we can skip question
 2 from the numbering.

Find the general solution of the equation

$$3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = x^2 \quad (3.1)$$

The auxillary equation is

$$3\lambda^2 + 4\lambda + 1 = 0 \quad (3.2)$$

which factorises to

$$(\lambda + 1)(3\lambda + 1) = 0 \quad (3.3)$$

and so has solutions

$$\lambda = -1 \text{ and } \lambda = -\frac{1}{3} \quad (3.4)$$

As both roots are real and distinct, the complementary function is

$$y_c = Ce^{-x} + De^{-\frac{1}{3}x} \quad (3.5)$$

Now, let us find the particular integral. As the right hand side of equation 3.1 is x^2 , our trial solution is the polynomial

$$y_p = px^2 + qx + r \quad (3.6)$$

$$\Rightarrow \frac{dy_p}{dx} = 2px + q \quad (3.7)$$

$$\Rightarrow \frac{d^2y_p}{dx^2} = 2p \quad (3.8)$$

Substituting the trial particular integral into equation 3.1

$$6p + 8px + 4q + px^2 + qx + r = x^2 \quad (3.9)$$

$$\Rightarrow px^2 + (8p + q)x + (6p + 4q + r) = x^2 \quad (3.10)$$

$$\Rightarrow p = 1, \quad q = -8, \quad r = 26 \quad (3.11)$$

Thus the particular integral is

$$y_p = x^2 - 8x + 26 \quad (3.12)$$

and combining equation 3.5 with equation 3.12, by the rule of superposition, we get the general solution of equation 3.1 to be

$$y = Ce^{-x} + De^{-\frac{1}{3}x} + x^2 - 8x + 26 \quad (3.13)$$